



UNIVERSITY OF CALOOCAN CITY
COMPUTER ENGINEERING DEPARTMENT



Software Design

Progress Report No. 8

Data Structures and Algorithms

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I. Objectives

In this section, the goals in this laboratory are:

- To describe a good conceptual design of the system
- To identify one of the three levels of structure of program – Data Structure

II. Methods

One of the most difficult things for program readers to understand is the way in which data is structured and used. A data map is very useful in interpreting the code's actions, especially when the system handles many files of varying types and purposes.

Why do we need Data Structure?

- It gives different level of organizing data
- It explains how data can be stored and accessed
- It provides operation on group of data, such as adding an item, inserting an item, sorting items, deleting an item, and updating items
- It provides a mean to manage huge amount of data efficiently
- It provides fast searching and sorting a data

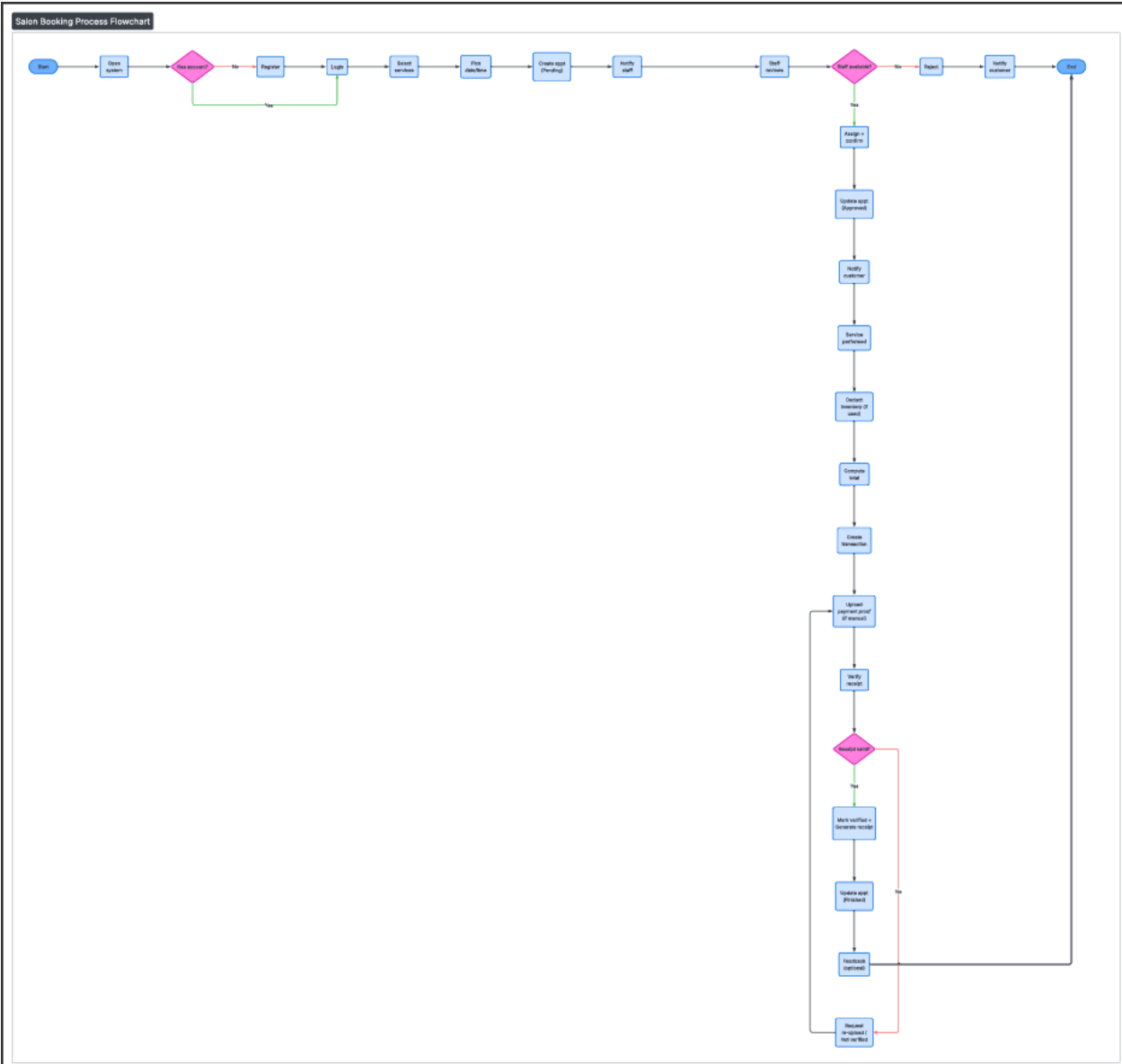
General Instructions:

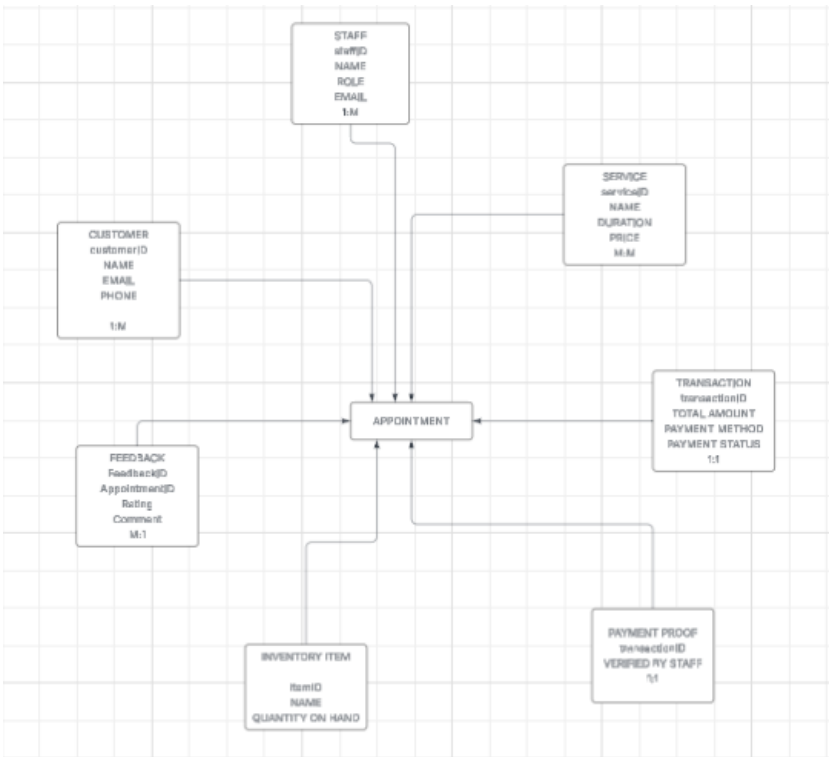
1. Create an Algorithm and Flowchart, Data Flow Diagram/ERD of your proposed system (Use LucidChart to draw your flowchart and other drawing software)

Algorithm

2. Start.
3. Customer opens the salon booking system.
4. Customer logs in or registers an account.
5. Customer selects service(s).
6. Customer chooses preferred date and time.
7. System creates an appointment with **Status = Pending**.
8. System notifies staff about the new pending appointment.
9. Staff reviews the appointment request.
10. If staff is **not available**, staff rejects the appointment and the system notifies the customer. End.
11. If staff is **available**, staff confirms the schedule and assigns a staff member.
12. System updates appointment **Status = Approved** and notifies the customer.
13. Customer goes to the salon on the scheduled date/time.
14. Staff performs the selected service(s).

- 15. If inventory tracking is used, system deducts used inventory items.
- 16. After service is finished, system computes total price based on services.
- 17. System creates a transaction record.
- 18. Customer uploads payment proof (receipt screenshot) if manual verification is required.
- 19. Staff verifies the uploaded receipt.
- 20. If receipt is **invalid**, system marks payment as not verified and requests re-upload. Go back to step 17.
- 21. If receipt is **valid**, system marks payment as verified.
- 22. System generates and stores the receipt.
- 23. System updates appointment **Status = Finished**.
- 24. Customer may submit feedback and rating.
- 25. End.





1. Customer creates an appointment to the salon.
The customer enters their details and requests a service by setting a preferred date and time.
26. Appointment is sent to the staff.
The system forwards the appointment to available staff members, and one staff is assigned to handle it.
27. Staff manages the appointment.
The staff checks the request, communicates with the customer if needed, and approves the final schedule.
28. Appointment is linked to services.
An appointment can include many services, and each service can be used in many appointments.
29. Inventory items are used during the service.
While doing the service, staff may use inventory items and these are deducted from stock.
30. After the service, a transaction is created.
Once the appointment is finished, the system generates a transaction containing the total amount and payment method.
31. Customer completes the payment.
The customer pays, and the transaction status is updated in the system.
32. Customer gives feedback.
After everything is done, the customer can submit feedback and rating for that appointment.

The data structure of the system consists of different entities such as Customer, Staff, Appointment, Service, and Transaction. These structures are used to store and manage information needed for booking, scheduling, payment processing, and feedback. Each data structure contains specific fields that represent the attributes of the entity.

II. Conclusion

In this project, I was able to design a conceptual model, algorithm, and flowchart for the Web-Based Salon Management and Booking System. This system helps replace the manual way of handling salon appointments by making the process easier and more organized. I learned how customers can book appointments online, how staff can manage schedules, and how transactions and feedback are recorded in the system. Through this activity, I understood the importance of proper system design and data structure in creating an efficient and reliable information system. Overall, this project helped me better understand how a real-world system works from start to finish

References

[1] I. Sommerville, Software Engineering, 10th ed. Boston, MA, USA: Pearson, 2016.